

EV DISCUSSIONS ON BLUESKY

Text mining, engagement testing, clustering and reply-network analysis

COMP3020 • GROUP 9 (Ishan , thevindu , sachini)

How do people discuss electric vehicles, which topics attract likes, and how are reply conversations connected?

3,976

usable posts

1,240

“charg” mentions

p = 0.0015

category–like association

1,748

accounts in reply network

DATA & APPROACH

In R, the following searches were used to collect posts about bluesky: electric vehicles, electric cars, Tesla, and EV charging. There were 3,976 usable posts that were cleaned up. Analyses: Word frequency/Word cloud, chi-square test of independence, K-means clustering on 463 posts and directed reply-network centrality.

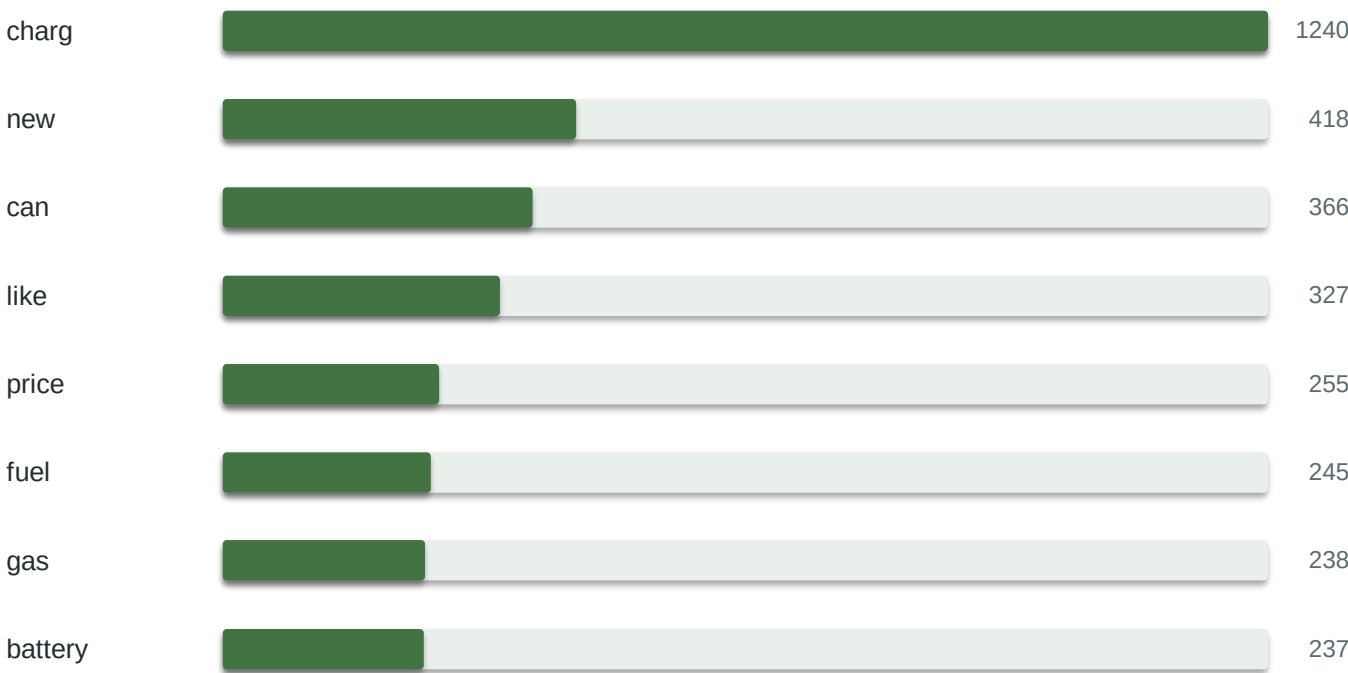
1 WHAT ARE PEOPLE DISCUSSING?

The most common term related to the topic was “charg” (1,240 times). Other recurring terms included price (255), fuel (245), gas (238), battery (237) and drive (230).

INTERPRETATION

Affordability, energy use and practical EV ownership are prominent, along with charging infrastructure. There is no correlation between word frequency and post support or opposition to EVs.

MOST FREQUENT TERMS



2 WHAT ATTRACTS LIKES?

A chi-square test showed a significant relationship between whether likes were above the median and search category ($\chi^2 = 15.41$, $df = 3$, $p = 0.001498$).

Above-median = more than 1 like, Median = 1 like. This is not evidence that the search topic is causing engagement, this is an association.

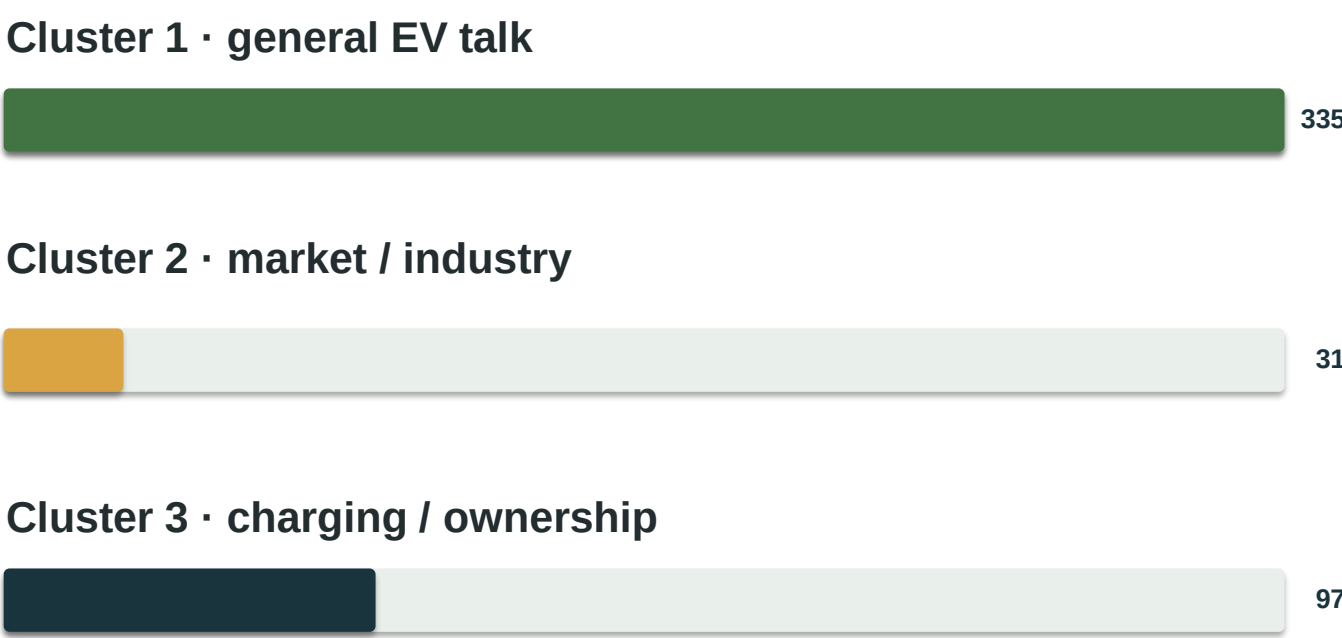
SHARE WITH ABOVE-MEDIAN LIKES



3 CAN POSTS BE GROUPED?

The three exploratory clusters obtained by the K-means procedure encompassed 463 posts. The elbow plot did not have a strong ‘elbow’ result and the MDS projection had overlap; thus the clusters are not well defined.

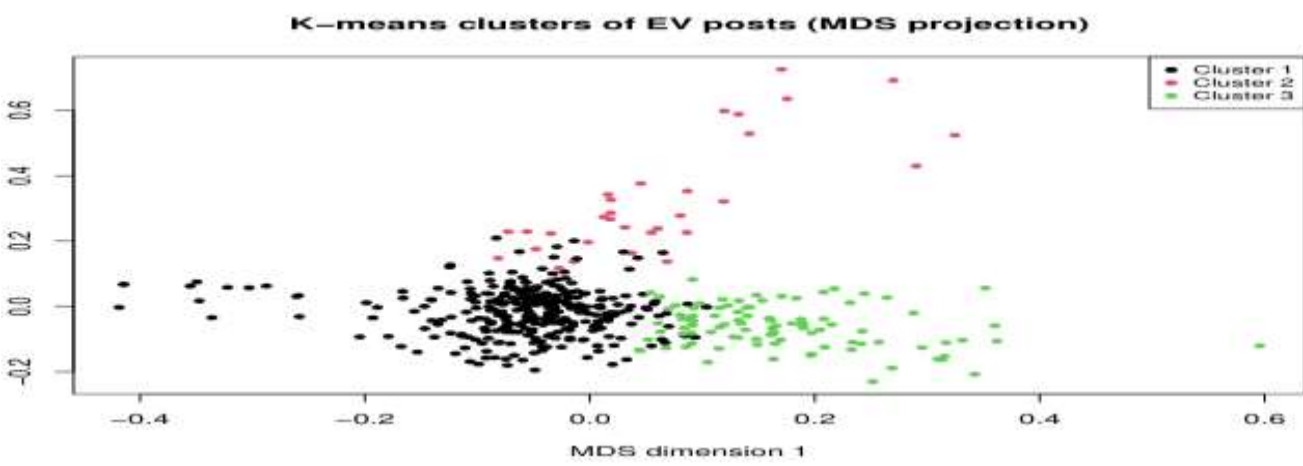
CLUSTER SIZES (n = 463)



Main themes from keywords:

- Everyday EV opinions and buying
- Market news and manufacturers
- Charging stations, home charging and cost

K-MEANS CLUSTERS: MDS PROJECTION



463 posts; themes are not completely distinct, rather exploratory, in the sense that there is an overlap of themes. Data from Group 9 report, page 8.

4 HOW ARE REPLIES CONNECTED?

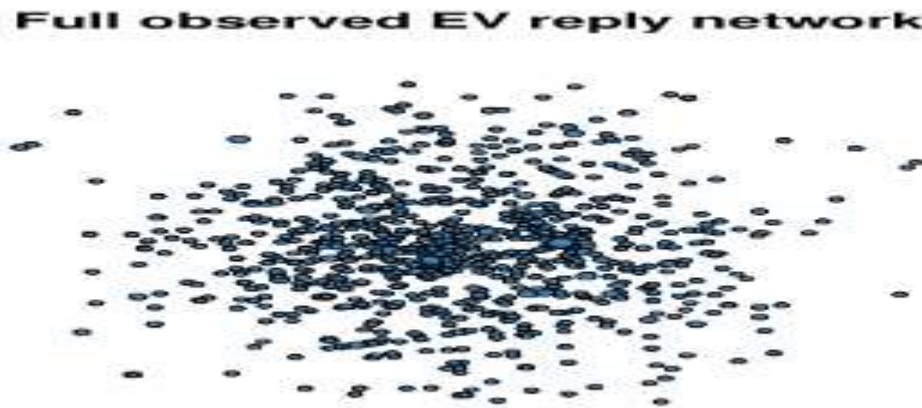
A total of 1,748 accounts and 1,141 directed reply ties were observed. It has a very low density (0.000374) and 636 weakly connected components, which means that the conversations are highly fragmented.

NETWORK SNAPSHOT

Accounts / nodes	1,748
Reply ties / edges	1,141
Network density	0.000374
Weak components	636
Largest component	50 nodes · 52 edges

Top observed account: in-degree = 21; PageRank = 0.00782. The account with the highest reported betweenness (53) was a separate account, which may have a bridging role..

OBSERVED EV REPLY NETWORK



The structure is sparse and dispersed, indicating broken up conversations that were sampled. Data from Group 9 report, p. 11.

OVERALL CONCLUSION

The themes of charge and cost and ownership features prominently in Bluesky EV conversations. The number of likes is also significantly varied by search category, with the posts about “Electric Cars” having the highest number of median likes (37.9%). Clustering highlights general but related topics and reply-network analysis indicates that conversations are spread out in numerous separate conversations..

LIMITATIONS

- Relevant chats may be excluded due to a collecting period and four search terms..
- Sentiment is not measured by word frequency.
- Clustering had no obvious elbow and only used 463 posts.
- Only responses were included in the network analysis; results cannot be applied to all Bluesky users..

NEXT STEPS

Extnded the sample and collection duration , incorporate sentiment analysis , and monitor the evolution of EV themes and reply networks over time .